

Evaluating design-stage manufacturability decisions against the resolution of production

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Abstract

Design-stage manufacturability decisions increasingly rest on simulation and machine learning, whose fitness to decide is rarely evaluated against production. A framework is proposed whose unit, the production floor, is the difference that drift and scatter of production and measurement produce between two batches of one design within a run. For each relation of a new design to the produced evidence, a rule returning meets, fails or trial has a decisive distance in floors, beyond which at least 95 % of its verdicts are correct, and a safe distance, beyond which at most 5 % are wrong; a procedure adds guard bands for single verdicts. Fourteen rules were evaluated at 95 % conformance on open data of 18 deep-drawn alternatives of 500 parts, mostly process settings, with matched simulations; a draw-in floor is 0.05 mm. With a produced sibling, the nearest produced setting was right for requirements over 3.4 floors from the truth (0.85 for near-replicates, 4.9 for resolvable siblings). For a new setting the best rule needed 4.7 floors interpolating, 4.9 extrapolating to 500 kN and 8.9 to 100 kN, with a stroke-speed change; for a new family the best was safe at 16–17 floors in either family's floor. Release-level requirements lay inside every decisive distance except a near-replicate's. With six to nine produced alternatives at two or three forces, the simulation, offset or rescaled, helped only across families, and learned 90 % intervals covered 85–96 % with a sibling, under 50 % extrapolating or across families. The framework gives evaluations of AI-based manufacturability support a production-referenced standard.

Keywords: design for manufacturing, machine learning, design decision making, verification and validation, operating characteristic, deep drawing

1 Introduction

Whether a design alternative can be made to its requirements is decided long before it is made. In sheet-metal forming ~~, as in most processes,~~ the decision rests on a finite-element simulation (Banabic, 2010): the team compares the predicted characteristics with the requirements and releases the alternative, changes it or sends it to a physical trial. Simulation has moved such decisions forward (Thomke, 1998), and machine learning makes the loop faster: surrogates replace simulations, models learn manufacturability from earlier designs, and checks are automated (Yan et al, 2026; Ghadai et al, 2018; Zhang et al, 2018). Neither has changed the question the team has to answer: *can this prediction be relied on to decide?* ~~A simulation that is right on average can be wrong for the alternative at hand, and a model learned on earlier products may be asked about a design unlike any of them.~~ Whitney argued in this journal that mechanical design cannot be verified by simulation the way integrated circuits are (Whitney, 1996); and production variation, with which development organisations still contend (Thornton et al, 2000), adds a further reason. ~~Whether:~~ whether a simulation-based decision is credible is ~~in the end~~ a property of production, and production evidence is ~~what is~~ scarce at the design stage.

This paper treats the design-stage manufacturability decision itself as the object of study. How can the fitness of a design-stage prediction ~~for such a decision to decide~~ be expressed on a common production-referenced scale, for any predictor and characteristic (RQ1)? How does it depend on the amount of production evidence and on how the new design relates to what has been produced (RQ2)? And what do simulation and machine learning contribute once production evidence exists (RQ3)? Decision-based design (~~Hazelrigg, 1998; Lewis et al, 2006~~) (Hazelrigg, 1998), design margins (~~Eckert et al, 2019; Brahma and Wynn, 2020~~) (Eckert et al, 2019) and the planning of verification and testing (Thomke and Bell, 2001; Salado and Kannan, 2018; Tahera et al, 2019) all need to know how far a prediction can be relied on ~~before a test is cheaper than a decision.~~

The framework proposed here has the *production floor* as its unit: the difference that drift and scatter produce between two batches of ~~consecutive parts of~~ one design within a run, a floor in the sense of a noise floor ~~rather than of the shop floor~~. A rule that turns the design-stage evidence into a verdict (the alternative *meets* the requirement, *fails* it, or goes to a physical *trial*) is characterised ~~by two operating characteristics in floors, its decisive distance and its safe distance,~~ for each *evidence relation* between a new design and the produced alternatives, by two points on its operating characteristics in floors, its decisive and safe distances, and a procedure turns them into a guard band for single verdicts. The decisions in scope are those of a team that has produced variants of a design family, in process planning and tryout and when production knowledge is carried to a new family; ~~early product design is covered only by the new-family relation.~~

The evaluation uses open data: 9,000 deep-drawn and cut cups of dual-phase steel, produced as 18 design-process alternatives of 500 consecutive parts each (Baum and Heinzelmann, 2026), with finite-element simulations of the same process (Baum and Heinzelmann, 2025). Within each of two cup geometries the alternatives differ in blank-holder force and lubrication on one tool, so two of the three relations concern process

settings, and the geometries give two transfers between families. Each alternative is judged as a new design, and the verdicts of thirteen by fourteen rules at a 95 % conformance level are compared with, against what its 500 parts did. In the terms of the design research methodology (Blessing and Chakrabarti, 2009), this is a prescriptive study with an initial support evaluation: it shows whether the framework discriminates between rules and evidence situations, not whether design teams decide better with it, and its findings on one dataset are hypotheses for confirmation.

The After the positioning (Section 2), the contributions are the framework (Section 3); an evaluation design that separates the evidence relations by grouped cross-validation (Roberts et al, 2017) and isolates what the simulation adds (Section 4); the empirical answers for deep drawing (Section 5); and guidance and a reporting standard for evaluating simulation- and AI-based manufacturability support (Section 6). Section 2 positions the work, Section 7 concludes, and Appendix A gives the notation, the robustness results and computation details.

2 Background and related work

2.1 Manufacturability evaluation and machine learning

Design for manufacturing (DFM) codifies what a process can make into rules, guidelines and cost models (Boothroyd et al, 2010), and quantitative manufacturability models have been used driven design decisions in this journal to drive design decisions (La Trobe-Bateman and Wild, 2003). Machine learning learns manufacturability from CAD models (Ghadai et al, 2018; Zhang et al, 2018; Williams et al, 2019) (Ghadai et al, 2018; Zhang et al, 2018), from process data and simulated design spaces (Yan et al, 2026), and generative models propose designs directly (Regenwetter et al, 2022); transfer between processes and product families remains difficult (Safdar et al, 2024). Learned manufacturability models are typically evaluated by held-out accuracy on labelled designs (Ghadai et al, 2018; Zhang et al, 2018; Williams et al, 2019), and language and vision-language models by case studies (Makatura et al, 2024) and benchmark questions (Picard et al, 2025), not by the decisions they support against the production of the alternatives they are asked about.

2.2 Design decisions, design types and production evidence

Decision-based design casts design as a choice between alternatives with uncertain outcomes (Hazelrigg, 1998; Lewis et al, 2006) (Hazelrigg, 1998). How much a team knows about a new alternative depends on the kind of design, variant, adaptive or original (Pahl et al, 2007), organised across products by product family and platform design (Simpson et al, 2006; Jiao et al, 2007). The platform design (Simpson et al, 2006); the evidence relations of this paper are their counterparts for production evidence: a new variant of a produced setting, a new process setting and a new family, the design type concerning the novelty of the solution and the evidence relation the novelty of its realisation relative to what has been produced. Production knowledge enters design through process-capability data (Thornton, 2004; Tata and Thornton, 1999) and

tolerance analysis (~~Chase and Parkinson, 1991~~). Unresolved uncertainty is absorbed by margins (~~Eckert et al, 2019; Brahma and Wynn, 2020~~), and the part of a margin (~~Eckert et al, 2019~~), whose part that covers uncertainty is distinct from the part that absorbs change (~~Eckert and Isaksson, 2017~~). Tests and prototypes, the alternative to a margin, are modelled as a trade between the costs of tests and of errors (~~Thomke, 1998; Thomke and Bell, 2001; Salado and Kannan, 2018~~) and studied as design activities (~~Tahera et al, 2019; Camburn et al, 2017~~) (~~Tahera et al, 2019~~). When upstream information is fit to act on is the question of overlapped development (~~Krishnan et al, 1997; Terwiesch et al, 2002~~), and set-based concurrent engineering codifies production knowledge as feasibility limits ~~for new designs~~ (~~Sobek et al, 1999~~). All of these assume a measure of how far a design-stage prediction can be relied on; the framework supplies one in a unit set by production.

2.3 Simulation credibility, calibration and validation

Verification and validation ask whether a model is accurate enough for its intended use, and the distinction between its validation and application domains (~~Oberkampf and Roy, 2010~~) is the question that the evidence relation asks of a design, and risk-informed credibility assessment scales the required evidence with the consequence of a wrong decision (~~ASME, 2018~~). Bayesian calibration adds a discrepancy model and calibration parameters to a simulator (~~Kennedy and O'Hagan, 2001~~), multi-fidelity models scale a cheap model to an expensive one (~~Kennedy and O'Hagan, 2000~~), and ~~omitting the discrepancy biases the parameters~~ (~~Brynjarsdóttir and O'Hagan, 2014~~); where a model may be relied on ~~at all is the question of only within~~ its valid input domain (~~Malak and Paredis, 2010~~). Conformal prediction gives distribution-free intervals ~~with finite sample coverage~~ under exchangeability (~~Lei et al, 2018; Barber et al, 2021~~), also under covariate shift (~~Tibshirani et al, 2019~~) and for ~~the control of~~ decision risks (~~Angelopoulos et al, 2024~~). In sheet-metal forming, friction (~~Hol et al, 2012~~) and the unloading behaviour of the material (~~Yoshida and Uemori, 2002~~) are ~~among the main sources of simulation error~~; simulation datasets of deep drawing support surrogate and transfer learning (~~Baum and Heinzelmann, 2025; Heinzelmann et al, 2025~~), and the deviation between simulation and parts has been decomposed statistically (~~Baum et al, 2026~~). How close to a requirement such intervals and accuracy statements can be relied on to decide, and how much production evidence that takes, remains open.

2.4 Decisions with abstention and their operating characteristics

In conformity assessment a measured value proves conformance, proves non-conformance or leaves the decision open, with guard bands set by the measurement uncertainty and the risks of false acceptance and rejection as the measures of a decision rule (~~ISO, 2017; Joint Committee for Guides in Metrology, 2012~~) (~~ISO, 2017~~). In selection procedures and acceptance sampling the operating characteristic gives the probability of a decision against the true quality, with an indifference zone inside which no decision is required (~~Bechhofer, 1954~~). A classifier with a reject option trades

errors against abstentions (Chow, 1970; Geifman and El-Yaniv, 2017), learning to defer passes difficult cases to an expert (Madras et al, 2018), and appropriate reliance on automation requires that its reliability be visible to its users (Lee and See, 2004), also in design teams working with AI (Zhang et al, 2021). The decisive and safe distances are operating characteristics of this kind, and the ~~guard band of the procedure~~ procedure’s guard band is the design-stage counterpart of a conformity-assessment ~~guard band~~ done.

2.5 Production variation and measurement

Statistical quality control compares short- and long-term variation with tolerances through capability and performance indices (Montgomery, 2019), with release criteria such as a performance index of at least 1.33 (Automotive Industry Action Group, 2006) and time-dependent models for drifting processes (ISO, 2026). Precision experiments define the repeatability limit, the difference that two results under repeatability conditions exceed with a probability of 5 % (ISO, 2025), and intermediate precision under conditions that differ in time, operator or equipment (ISO, 2023); measurement systems analysis separates repeatability from stability (Automotive Industry Action Group, 2010). The production floor is a ~~repeatability-type limit of batch centres under within-run conditions~~ precision limit of this kind for batch centres within a run (Section 3.2); what is new is its use not to judge whether two results agree but as the unit of in which a design decision rule is scored and its guard band set. Table 1 places the framework among ~~these approaches~~ them: it does not replace calibration or conformal prediction but scores them ~~as~~ as decision rules ~~against~~ against production in a common unit.

3 The framework

3.1 The decision problem and its scope

A design alternative a is a combination of product geometry and process settings ~~that is~~ produced as a series of parts. ~~A~~ over which a quality characteristic c varies ~~over the parts of a series~~. A requirement is an upper limit R on the characteristic, or on its absolute value for characteristics whose target is zero, that a share p of the parts must meet: alternative a is *adequate* for c when the p -quantile of its parts satisfies $q_p(a, c) \leq R$. Lower or two-sided limits follow by a change of sign or by the absolute deviation from the target; ~~a tolerance of $\pm t$ on a wall angle about its design angle, for example, is the limit $R = t$ on the absolute deviation~~. At the design stage q_p is unknown. A decision rule r maps the available evidence to an interval $[L_r, U_r]$ for q_p and to a verdict

$$V_r = \begin{cases} \text{meets} & U_r \leq R, \\ \text{fails} & L_r > R, \\ \text{trial} & \text{otherwise.} \end{cases} \quad (1)$$

The three verdicts mirror the decision rules of conformity assessment (ISO, 2017), here for the uncertainty of a design-stage prediction. A rule that returns a point ($L_r = U_r$) always decides. The evidence consists of simulations of a and of the production record

Table 1 Positioning among methods that support or evaluate design-stage manufacturability decisions

Approach	Evidence	Output	Scored against production	Abstains
DFM rules and cost models (Boothroyd et al, 2010)	codified process knowledge	guideline compliance, cost	no	no
Design margins (Eckert et al, 2019)	experience, analysis	buffer to the requirement	rarely	no
V&V, credibility assessment (Oberkampf and Roy, 2010; ASME, 2018)	physics model, validation tests	accuracy in physical units; required evidence	on test cases	can require tests
Bayesian calibration (Kennedy and O’Hagan, 2001)	simulations, experiments	predictive distribution	not as decisions	implicitly
Conformal prediction (Lei et al, 2018)	exchangeable calibration data	interval with coverage	coverage; per decision possible on the measured item	yes
Conformity assessment, OC curves (ISO, 2017; Bechhofer, 1954)	measurement and its uncertainty	accept, reject, undecided; risks		yes
Selective classification (Chow, 1970; Geifman and El-Yaniv, 2017)	labelled data	class or reject; risk–coverage	on held-out data	yes
ML manufacturability models (Ghadai et al, 2018; Yan et al, 2026)	historical or simulated designs	class or value	on held-out data	rarely
This paper	simulations, produced alternatives	verdict; decisive and safe distance in floors by evidence relation	yes, per verdict	yes

of other alternatives, the *calibration alternatives*; ~~it is used by the process planner or tryout engineer for a further variant or process setting of an existing tool, at tool tryout or before production part approval, and by the product designer for a new family.~~ The demonstration decides on $p = 0.95$, which for a normal characteristic corresponds to a performance index of about 0.55. Release criteria of series production, a performance index of 1.33 or more (Automotive Industry Action Group, 2006), concern nonconforming fractions in parts per million that a series of 500 parts cannot estimate without a parametric tail; ~~the framework reaches them through requirements anchored in capability~~ (Section 5.8) locates such limits relative to the distances, without deciding them.

3.2 Production floor, effect-to-scatter ratio and minimum resolvable change

Production evidence comes as series of consecutive parts, each cut into batches of B consecutive parts, and each batch is summarised by a robust centre m (here the 10%

trimmed mean). The production floor of characteristic c is

$$F_c = Q_\alpha \{ |m_i - m_j| : i < j \text{ batches of the same alternative} \}, \quad (2)$$

the α -quantile of the difference between two batches of one design. Under a stationary normal model with a standard deviation σ_b between batch means and σ_w between parts, $F_c = 1.96\sqrt{2}\sqrt{\sigma_b^2 + \sigma_w^2}/B$ at $\alpha = 0.95$; ~~the form of~~ the repeatability limit of precision experiments ~~(ISO, 2025, 1994), applied to batch centres under within-run conditions~~ (ISO, 2025). Because it pools batch pairs up to 450 parts apart and grows with their lag (Section 5.1), it is closer to a time-different intermediate-precision limit (ISO, 2023) of batch centres within a run; in the terms of ISO 22514-2 the runs follow a time-dependent model with varying location (ISO, 2026). The reference protocol of this paper is $B = 50$, ten batches of a 500-part run, all pairs of batches and $\alpha = 0.95$, with per-geometry floors where geometries differ. The floor rests on three assumptions: batches of one design differ only by drift and scatter within a run (A1); the variation within the runs from which it is estimated ~~are like those of future production is like that within future runs~~ (A2); and the measuring system is repeatable and its measurement resolution small against the floor (A3, checked in Section 5.1).

Four properties matter for its use. It has the unit of the characteristic, so distances in floors are on a common scale for all characteristics and predictors, ~~although though~~ equal distances do not mean equal nonconforming fractions ~~because the ratio of the floor to the scatter of the parts differs between characteristics~~ (Table 3). It is a property of a ~~production~~ protocol: under drift it grows with the time between the batches compared ~~and it and~~ shrinks with B ; ~~and~~ σ_b and σ_w are reported so that it can be recomputed ~~for another protocol~~ (Table 3). It contains whatever drifts between batches, ~~the process and the process and~~ measuring system alike, so without a reference artefact scanned through the series it is a production-and-measurement floor. And it excludes the variation between runs, which makes it a lower bound of the variation of series production. ~~For a A~~ new family, whose floor is unknown before it is produced, ~~is scored in~~ the floor of the produced family ~~is used. Because, and because~~ the floor resolves batch centres ~~while the requirement concerns a quantile~~, the reproducibility of the ~~decided~~ quantile is reported alongside.

Two alternatives a and b are resolvable for c when their effect-to-scatter ratio

$$\text{ESR}_c(a, b) = |\bar{m}_a - \bar{m}_b| / F_c \quad (3)$$

is at least one, $\bar{m}$ being the centre of all parts of an alternative; this is a practical threshold, not a test. For a continuous design or process parameter x with a local sensitivity $s_c = \partial \bar{m} / \partial x$, the minimum resolvable change is $F_c / |s_c|$: a change of x by less moves the centre of the characteristic by less than two batches of one design differ. It is local ~~a statement about centres, and and concerns centres~~; a change below it can still move the tail ~~of the distribution~~.

3.3 Decision rules

~~Thirteen-Fourteen~~ rules span what a design team can do with its evidence (Table 2);
~~each is defined for any set of calibration alternatives, and~~ the first seven carry the
~~results in the main text. The nominal simulation main results. Beside the nominal~~
~~simulation s_0 (M0) uses the simulation alone. The,~~ the bias-corrected simulation (M1)
adds the median offset between the calibration alternatives' q_p and their ~~nominal~~
~~simulation s_0~~ ; the scaled simulation (M1s) fits $q_p = a + b s_0$ to them by least squares,
a linear multi-fidelity correction (Kennedy and O'Hagan, 2000) that rescales a simu-
lated trend of the wrong size. The envelope rule (M2) adds the median offset and the
largest absolute residual of the calibration alternatives to the upper end of the simu-
lation envelope over plausible incoming conditions, which vary sheet thickness and
friction but not material properties. M5 models the offset between q_p and the enve-
lope as a Gaussian process (Rasmussen and Williams, 2006) with a 90 % predictive
interval, whose noise variance is bounded below by the variance between the ~~produced~~
~~calibration alternatives'~~ series of one geometry and force, so that a new series of a
produced setting is not predicted more precisely than produced series reproduce each
other; M5n is the same process of q_p without the simulation. The nearest produced
setting (NN) is a capability-database look-up that uses neither simulation nor learn-
ing. The ~~six-seven~~ further rules include a friction-calibrated simulation (Kennedy and
O'Hagan, 2001) (Mc)~~and,~~ part-level gradient boosting (Friedman, 2001) with a jack-
knife+ interval (Barber et al, 2021), with and without the simulation (M3, M3n), and
the least-squares line of q_p in force (M1sn). The nominal simulation and the envelope
do not depend on the lubrication pattern, so ~~a lubrication variant of a produced setting~~
~~within a geometry they are functions of the force alone: M1sn is the counterpart of~~
~~M1s without the simulation, and a lubrication variant~~ gives M0, M1, M1s, M2 and
M5 no simulated information that distinguishes it from its siblings.

~~The margins are labelled by what they guarantee.~~ With n calibration alternatives,
the largest-residual margin covers a new exchangeable residual with probability $n/(n + 1)$
for a fixed offset and at least $(n - 1)/(n + 1)$ for an estimated one, ~~because the new~~
~~residual is then covered whenever it falls within the range of the calibration residuals;~~
it is the split-conformal quantile (Lei et al, 2018) at the highest attainable level, not
a 90 % interval. ~~In general, a~~ A distribution-free interval cannot certify a miss rate
below $1/(n + 1)$, so a miss rate of 5 % needs at least 19 produced alternatives in the
relevant relation ~~and one of 10 % needs nine. Under exchangeability,~~ With six to nine
~~calibration alternatives the jackknife+ interval covers with probability at least 0.8~~
~~bounds at the level of 0.9 used here (Barber et al, 2021), and the fall at or beyond~~
~~the extreme leave-one-out values, which are used instead; under exchangeability these~~
~~guarantee a coverage of $1 - 2/(n + 1)$, 0.71 to 0.80 (Barber et al, 2021), approximately~~
~~because the residuals are centred on their median. The~~ Gaussian-process interval is
model-based. The level ~~is a decision variable that~~ trades decisiveness for safety, as the
cost of errors would set the threshold of a reject option (Chow, 1970); 0.9 is used here,
~~and 0.8 and 0.95 are reported~~ in the robustness analysis.

Table 2 The decision rules; the first seven carry the results in the main text

Rule	Prediction of q_p	Interval	Evidence
M0 nominal simulation	simulation at nominal thickness and friction	point	none
M1 bias-corrected simulation	M0 plus the median offset	point	q_p
M1s scaled simulation	$a + b$ M0, least squares	point	q_p
M2 envelope + residual margin	envelope top plus the median offset	largest residual	q_p
M5 GP-corrected envelope	envelope plus a GP offset over the descriptors	GP, 90 %	q_p
M5n GP of q_p	GP of q_p over the descriptors	GP, 90 %	q_p
NN nearest produced setting	mean q_p at the nearest produced setting	point	q_p
M0w process-window worst case	simulated range over the incoming conditions	range	none
Mc friction-calibrated simulation	friction calibrated on the draw-in, plus offset	point	centres, q_p
M3 simulation + boosting	matched simulation plus boosted discrepancy	jackknife+	parts
M3n boosting, no simulation	boosted model of the characteristic	jackknife+	parts
M1n median of the produced	median q_p of the calibration alternatives	point	q_p
M1sn force trend	$a + b$ force, least squares	point	q_p
M2n median + residual margin	M1n with the largest-residual margin	largest residual	q_p

Fitted on the calibration alternatives. Descriptors: geometry indicator, blank-holder force and lubrication rank (the order of the oil amount). GP: ~~noise variance bounded below by the variance between the series of one geometry and force~~ Gaussian process. M3 and M3n at part level; M3n is named M4 in the released code; M6, a conformalised GP, ~~appears only in the~~ is a robustness analysis variant (Table 4).

3.4 Operating characteristics: decisive and safe distances

A rule is scored by judging each produced alternative as a new design at requirements placed at a known distance from the truth,

$$R = q_p(a, c) + d F_c, \quad (4)$$

so that a is adequate exactly when $d \geq 0$. For a case (a, c) with interval $[L, U]$, let $u = (U - q_p)/F_c$ and $v = (q_p - L)/F_c$ be the distances by which the interval excludes the truth from above and below. At $d = +\delta$ the verdict is correct when $u \leq \delta$ and a false reject when $-v > \delta$; at $d = -\delta$ it is correct when $v < \delta$ and a false accept when $-u \geq \delta$. Let $\kappa(\delta)$ and $\omega(\delta)$ be the shares of correct and of wrong verdicts among all feasible verdicts at $|d| = \delta$: every case gives one verdict on an adequate alternative and, where $q_p - \delta F_c \geq 0$, one on a failing alternative, and every verdict has equal weight. As δ grows, failing-side requirements become infeasible for characteristics close to zero, so the failing side carries half the weight up to a few floors and a third at about 100 floors (98 of 126 failing-side requirements remain feasible at 16 floors and

63 at 108). The *decisive distance* δ_d of a rule is the smallest δ beyond which $\kappa \geq 0.95$, and its *safe distance* δ_s the smallest δ beyond which $\omega \leq 0.05$.

The distances are exact up to the 0.05-floor grid on which they are evaluated: κ and ω change only where δ crosses one of the case's u , v , $-u$ or $-v$, so the distances are order statistics of these exclusion distances; for a rule that always decides, $\delta_d = \delta_s$ is close to the 90 % quantile of its absolute error in floors. Always $\delta_s \leq \delta_d$, and between the two the rule sends designs to trial rather than misleading; one-sided distances for failing alternatives (false accepts) and adequate ones (false rejects) matter when the errors cost differently. The distances are points on operating characteristics over a population of design situations: δ_d is the analogue of the indifference zone of a selection procedure (Bechhofer, 1954) at a probability of correct selection of 0.95, ~~and the pair plays the role of the acceptable and rejectable quality levels of an acceptance plan~~. A trial is ~~itself~~ not error-free: a short run reproduces the 95th percentile only to about one floor within a run (Section 5.1) and, being a run, also carries the variation between runs, so no decision ~~can be relied on~~ is reliable closer to the requirement than the trial ~~that would replace~~ replacing it.

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432 3.5 Evidence relation, calibration budget and applicability 433 guard 434

How a new design relates to the produced alternatives is part of the decision situation. With a *sibling*, an alternative at the same process setting has been produced ~~with another variation that that differs only in the factor varied at a fixed setting (here the lubrication pattern), which~~ production may not resolve; a *new process setting* lies between produced settings (interpolation) or outside them (extrapolation); a *new family* shares no produced alternative. ~~Beyond a factorial, the relation can be graded by a distance in descriptor space or by the effect-to-scatter ratio between the predicted characteristic and the nearest produced alternative (Section 6.5).~~ The calibration budget is the dependence of the distances on the number k of produced alternatives, over all subsets of size k classified by relation. An applicability guard returns *trial* when the new design lies outside the calibration: its geometry is not calibrated (family guard), a descriptor lies outside the calibrated range (range guard), or its simulated q_p lies outside the calibration alternatives' simulated values (novelty guard), in the spirit of a valid input domain (Malak and Paredis, 2010).

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450 3.6 Procedure

Figure 1 turns the framework into six steps: steps 1 to 5 once per design family, repeated when production evidence is added, and step 6 for every new design. ~~Step Steps 1 frames to 4 frame~~ the decision (family, alternatives, characteristics, form of the requirement ~~and~~ conformance level); ~~step 2 measures~~, measure the floor with a stated protocol; ~~step 3 states which differences between alternatives, and which local changes of a parameter, production can resolve; and step 4 calibrates~~, state which differences production resolves and calibrate the candidate rules with and without the simulation. Step 5 qualifies them by their distances for each evidence relation and fixes for each rule and relation a *guard band* g : the smallest observable margin

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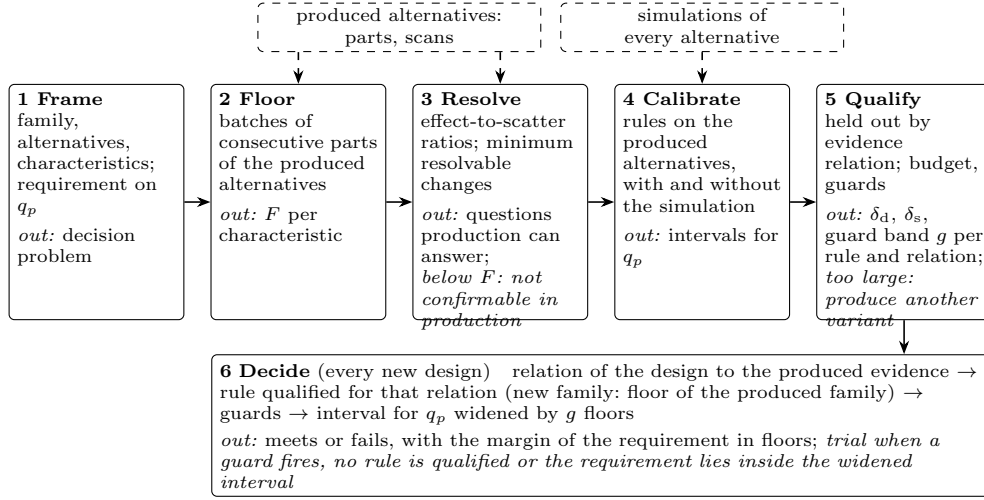


Fig. 1 The procedure: steps 1–5 once per design family, step 6 for every new design; dashed boxes are the evidence it uses

between the predicted interval and the requirement beyond which at least 95 % of the decided verdicts were correct. Being conditional on what is observed, g depends on the distribution of requirement distances that the team expects, not on rule and relation alone. The reference prior weights the 47 distances of the evaluation grid within ± 20 floors equally (0, for requirements within 20 floors of the truth ± 0.5 to ± 10 in steps of 0.5, ± 12 , ± 15 and ± 20 ; 87 % of the weight within ± 10), drops infeasible requirements and searches g in steps of 0.25 floors up to 20; it represents requirements set from a few to some tens of floors from the true 95th percentile, denser where verdicts are hardest. If the distances are too large for the requirements of coming designs, another variant is produced. Step 6 establishes the relation of the new design, applies the rule qualified for it and the guards, widens the rule's interval by g floors and returns meets or fails with the margin in floors, or a trial when a guard fires, no rule is qualified or the requirement lies inside the widened interval; Section 5.7 reports g under other priors and works one decision through.

4 Demonstration on deep drawing

4.1 Alternatives and production data

The production data are the Real Deep Drawing and Cutting dataset (RDDAC) of the University of Stuttgart (Baum and Heinzlmann, 2026; Baum et al, 2026). Square blanks of DP600 steel, 210 mm wide and about 1 mm thick, were deep drawn into cups (OP10) and cut (OP20) on one tool set with two geometries, concave and convex side walls, which differ in base curvature, wall angle and bottom radius at once. Each geometry was run at three blank-holder forces (100, 300 and 500 kN) and three lubrication patterns (coarse, medium and fine, with series medians of 1.15–1.48, 0.92–1.23 and 0.85–1.13 g/m² of oil): 18 alternatives, each produced as one series of 500

consecutive parts, the change of geometry being the only change of product design. Every part has press signals, sheet and oil-film traverses and 3D scans after both operations; ~~10 of the 9,000 parts carry no scan~~. The series warm up by 1.0–5.0 K over their first 150 parts from punch temperatures of 20.5–29.6 °C, their median sheet thickness ~~ranges from~~ is 0.986 ~~to 0.993~~ 0.993 mm, and the 100 kN series ran at 47 mm/s against 78–79 mm/s. The documentation states neither the order and dates of the series nor whether the parts were scanned in production order.

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515 4.2 Simulations

The simulations come from the Deep Drawing and Cutting Simulations dataset (DDACS) of the same group (Baum and Heinzelmann, 2025; Heinzelmann et al, 2025): LS-DYNA quarter models with nodes about 1 mm apart, a scaled DP600 flow curve, constant Coulomb friction and springback after both operations; the yield criterion and unloading model, which govern springback (Yoshida and Uemori, 2002), are not documented, and the stroke speed is not represented. For the RDDAC tools, 396 simulations vary the sheet thickness (0.95–1.00 mm) and the friction coefficient (0.05–0.15) at each geometry and force. A part is matched to the simulation with the nearest thickness and the friction coefficient that the dataset’s linear mapping assigns to its oil film, an assumption of the dataset’s authors on which Mc, M3 and every matched simulation depend. The nominal simulation of an alternative is at 0.99 mm, the grid value nearest the median thickness, and friction 0.10.

529

530 4.3 Quality characteristics and their measurement

Seven characteristics were extracted with the same definitions from the scans and the simulated nodes, referred to the plane of the cup bottom: the mid-side and corner draw-in and the flange waviness, process indicators on the flange that the cut removes, and four characteristics a drawing would carry: the wall angle (design angle 20° concave, 10° convex), the cup depth, and the arm angle and bottom dome of the cut part, the last two with requirements on absolute values. The four sides of a part measure several of them twice, which exposes the measurement: the flange width along the scan lines scatters 4 to 18 times more than across them, so the mid-side draw-in is taken across the scan lines, and the wall angle is the mean of the east and west walls ~~(the north wall carries uncorrelated scatter, the south wall is shadowed)~~. The mid-side draw-in is quantised in steps of 0.019 mm (0.38 floors) by the pixel pitch. Differences between scans and simulations, such as the scanned surface against the shell mid-surface, enter the offsets between the two but not the floors.

544

545 4.4 Evaluation design

Every produced alternative is judged as a new design under grouped cross-validation (Roberts et al, 2017), one scheme per evidence relation. *New variant*: the alternative alone is held out and the other eight of its geometry calibrate, two siblings at its force among them. *New process setting*: the three alternatives of one force of a geometry are held out and the six others calibrate; 300 kN is interpolated, 100 and 500 kN extrapolated. *New family*: one geometry calibrates the rules for the other (two

transfers). Pooled versions of the first two add the other geometry’s alternatives to the calibration set. Every calibrated rule is evaluated with and without the simulation (M1/M1n, M1s/M1sn, M2/M2n, M3/M3n, M5/M5n), ~~so that the paired difference isolates isolating~~ what the simulation contributes, and the nearest produced setting is the baseline without simulation and learning. A scope comprises 126 cases (18 alternatives, 7 characteristics).

The ~~distances are computed exactly. Their intervals~~ intervals of the distances come from 1000 resamples of the held-out alternatives within each geometry, each with a block-bootstrap replicate of every true q_p , common to all rules so that differences get paired intervals; they hold the fitted rules fixed. A second bootstrap ~~resamples whole lubrication patterns within each geometry~~ refits every alternative-level rule after drawing the three lubrication patterns of each geometry with replacement, at least two of them distinct, which keeps every case in its evidence relation ~~and refits every alternative-level rule~~ (Table A2). Its 200 resamples contain 48 of the 49 distinct configurations; in three of four draws a pattern is duplicated, so that a new variant keeps one distinct sibling and the duplicated series enter the noise floor of the Gaussian processes as identical pairs. Its intervals thus show the effect of losing a pattern, not the sampling uncertainty of the reported values.

4.5 Computation and development record

The constants of the floor and the requirement grid were fixed before the decision results were seen, as the repository history records. The rules were not: M0–M5 were defined first; M0w, Mc, M1n, NN, M2n, M5n, the grouped schemes, the exact estimator and the noise floor of the Gaussian processes were added after the first results; ~~and~~ a review of that version added M1s, the jackknife+ intervals, the between-run noise floor and the analyses of Sections 5.4 and 5.7; and a second review added M1sn, the guard bands under other priors and the new family in the produced family’s floor. The comparison of rules is therefore exploratory and carries the optimism of a selection made on the evaluation data (Cawley and Talbot, 2010); every variant tried is reported in the paper or in the archived results ~~The pipeline reproduces every result from the cached feature tables in about four hours on four cores~~ (Canseven, 2026), which Table A5 maps to the text.

5 Results

5.1 What the production floor contains

In 95 % of batch pairs, two batches of 50 consecutive parts of one alternative differ by up to 0.051 mm in mid-side draw-in and 0.078° in wall angle (Table 3); one floor is ~~0.66 to 2.2~~ 0.7 to 1.7 long-term standard deviations of the parts. The floors are set by drift within the run: without the time order they shrink by a factor of 2.1 to 4.5, and batches one position apart differ by 0.51–0.80 floors and nine apart by 1.1–1.5 ~~and the~~ (averages over the two geometries). The normal model of Section 3.2 reproduces the floors to within 12 %; for subgroups of five parts, as sampled inspection would

Table 3 Production floor per characteristic, pooled and per geometry, with its components

Characteristic	F	95 % interval	Concave	Convex	F/σ_{LT}	σ_b/F	σ_w/F	Drift
Draw-in, mid-side (mm)	0.051	0.045–0.061	0.050	0.052	1.0/1.4	0.32/0.34	0.91/0.66	3.3
Draw-in, corner (mm)	0.046	0.035–0.052	0.053	0.030	1.4/1.2	0.31/0.35	0.65/0.77	4.5
Flange waviness (mm)	0.0028	0.0024–0.0031	0.0037	0.0021	1.6/0.7	0.33/0.30	0.55/1.31	2.5
Wall angle ($^\circ$)	0.078	0.066–0.10	0.081	0.078	1.3/0.7	0.32/0.26	0.68/1.39	2.1
Arm angle, cut ($^\circ$)	0.11	0.094–0.12	0.12	0.100	1.7/0.8	0.36/0.31	0.46/1.22	2.6
Cup depth (mm)	0.011	0.0096–0.012	0.0087	0.012	1.4/1.6	0.31/0.34	0.65/0.52	3.9
Bottom dome, cut (mm)	0.0070	0.0061–0.0081	0.0034	0.0082	1.2/1.6	0.33/0.36	0.76/0.55	3.9

Batches of 50 consecutive parts, 95 % quantile of the difference of ~~batch-centrestheir 10 % trimmed~~ means; interval from 2000 ~~batch~~ resamples ~~of batches within series~~. Ratios per geometry (concave/-convex), ~~classical statistics pooled over the series as root mean squares~~: σ_{LT} , ~~median long-term~~ standard deviation of the parts ~~of a series~~; σ_b , ~~standard deviation~~ between batch means, corrected for $\sigma_w^2/50$; σ_w , within batches. Drift: floor over ~~the floor that~~ of randomly permuted series.

~~measure, it gives floors 1.1 to 2.1 times as large (median 1.3)~~. Independent part-to-part noise, scanner repeatability included, would produce 15–33 % of each floor, and the ~~measurement~~ resolution is small against the floor except for the mid-side draw-in; ~~assumption~~ A3 holds for repeatability and resolution, but the stability of the scanner cannot be checked without a reference artefact. Nor do the process signals identify the source of the drift: batch centres of press force, punch temperature, sheet thickness and oil film predict it out of fold with R^2 of -0.23 to 0.20 , ~~and removing the predicted drift changes the floors by -16 to $+12$ %~~. Variation between runs is not in the floor: the three lubrication series of one geometry and force differ by up to 1.2 (waviness) to 9.8 floors (dome), and a floor taken across them is 1.5 to 9.1 times the reference. ~~The~~ ~~Within a run the~~ decided statistic reproduces to about one floor ~~÷ (the 95th percentiles of two batches of 100 parts differ by 0.92–1.49 floors in 95 % of pairs)~~, so distances below about one floor are beyond what production ~~, or a trial, can confirm~~ ~~can confirm, and a trial run, which also carries the variation between runs, confirms less still.~~

5.2 What production resolves and how far the simulation misses it

All 63 geometry contrasts exceed the floor and blank-holder force is resolved in 105 of 126 contrasts, but lubrication only in 51 of 126 (42 with a bootstrap probability of at least 0.95), so a lubrication variant is often a near-replicate of its siblings. The minimum resolvable force change is 21–67 kN between 100 and 300 kN for most characteristics and lies beyond either force interval for the wall angle and the dome. The nominal simulation misses production by far more than a floor: its median offset reaches 109 floors, its force effect on the draw-in is 2.7 to 5.1 times the measured one, and the cut arms of the concave cups bend downwards in 89–100 % of the simulations but upwards in every part. Part of the offset is definitional: the cup depth differs by a median 0.90 and 0.95 mm, consistent with the scanned surface against the shell mid-surface, and removing this offset alone brings the decisive distance of the nominal simulation from 108 to 65 floors.

5.3 Decisions by evidence relation

Table 4 and Fig. 2 score the rules. The nominal simulation ~~needs requirements~~ decides at 108 floors ~~from the truth (bootstrap interval (101–115) before 95 % of its verdicts are right; it errs, erring~~ mostly by rejecting adequate alternatives ~~(safe distance 17 floors for false accepts, 109 for false rejects).~~

~~A new variant with a produced sibling. With the other eight alternatives of the family produced, every~~ Every calibrated rule improves on ~~the nominal simulation~~ it, and the simplest is the most decisive: the nearest produced setting, the mean q_{95} of the two siblings ~~at the same force~~, is right beyond 3.4 floors (2.0–4.5), 0.17 mm of mid-side draw-in. ~~The sealed simulation follows at 4.9 floors; the Gaussian processes decide at 7.2 floors and but~~ are safe below one floor; ~~the envelope rule is safe at 0.15 floors but sends a third of the designs to trial at 10 floors and decides only at 23. Bias-correcting the nominal simulation (M1, 15 floors) is worse than the median of the produced alternatives without it (M1n, 9.1).~~

~~A new process setting. With every alternative at the new design's force held out, all rules lose. The: the force trend and the~~ nearest produced setting ~~is right beyond decide at 8.2 and 8.5 floors (5.5–10.2);~~, the interval rules ~~decide (M2, M2n, M3, M3n, M5, M5n) only at 13–31 floors,~~ though they are safe at 5.9–16 ~~and at 10 floors send 18–37 % of the designs to trial. Interpolating in force is much easier than extrapolating (the nearest setting decides at 4.7 against 9.5 floors, and M2 and M5 are safe at 2.7 and 2.8 against 9.9 and 8.9 floors. The sealed simulation decides at 19 floors and), and the sealed simulation~~, extrapolating a fitted trend, falsely accepts failing alternatives as far as 164 floors from their limit.

~~A new design family. In the two transfers~~ Scored in the floor of the produced family, as the procedure prescribes, every rule without the simulation fails ~~(131–133 in the two transfers (130–135 floors), because nothing in one family's production record describes the other. With the simulation, the bias correction decides at 35 floors, 33 floors and M5 at 38–35, and M3 is safe at 17 floors, M2 at 49, and the envelope rule is the safest at 16 floors; no rule is safe closer than that. Scored 26 and M5 at 31; in the floor of the produced new family, which a design team would have, M5 decides at 35 and team would not have, M2 is safe at 31 floors and the envelope rule 16, M3 at 41 and 26. Sixteen floors are about 21 and M5 at 28. The best rule was thus safe at about 16–17 floors in either floor, some 0.8 mm of mid-side draw-in or 1.3° of wall angle, but two transfers do not resolve which rule is safest: the safe distance of M3 minus that of M2 lies between –16 and +7 floors in the produced family's floor and between –3 and +8 in the new family's.~~

5.4 Where the distances come from

Three features of the ~~evaluation design shape the design shape these~~ distances. First, the sibling relation mixes replication with variation: in 58 of the 126 ~~new-variant~~ cases neither sibling differs resolvably from the held-out alternative, and the nearest ~~produced~~ setting is then right beyond 0.85 floors, the reproducibility of q_{95} between near-replicate runs; ~~when both differ it needs 2.5 floors when one sibling differs~~ resolvably (34 cases) ~~, it needs and 4.9 floors when both do (34 cases), where the force trend~~

Table 4 Decisive and safe distances (floors) by the relation of the new design to the produced evidence

Rule	New variant		New process setting				New family		Pooled	
	δ_d/δ_s	FA	δ_d/δ_s	FA	interp.	extrap.	δ_d/δ_s	FA	variant	setting
M0	108	17	108	17	109	106	104	28	108	108
M1	15	15	24	24	11	25	33	35	27	28
M1s	4.9	4.7	19	164	19	19	121	125	15	20
interval	4.7–6.3		14–21		19–20	15–19	119–121			
M2	23/0.15	0.30	18/8.1	8.3	18/2.7	18/9.9	41/26	27	53/0	53/1.0
interval	22–23		17–18		15–18	17–19	39–42/9.8–32			
M5	7.2/0.75	0.15	19/6.8	7.6	19/2.8	19/8.9	35/31	33	11/0.05	28/5.4
interval	6.6–8.4		18–20		18–20	18–20	31–39/28–32			
M5n	7.2/0.55	0.10	17/6.1	9.0	16/0	18/8.6	133/128	127	9.8/0.20	16/2.7
interval	6.4–7.5		16–18		15–16	16–19	131–133/127–128			
NN	3.4	2.8	8.5	10	4.7	9.5	130	129	3.4	8.5
interval	2.0–4.5		5.5–10		4.3–5.1	6.6–11	129–131			
M0w	119/94	9.2	119/94	9.2	119/94	119/77	117/79	10	119/94	119/94
Mc	27	28	39	39	14	40	43	53	32	33
M3	23/0	0	31/16	13	29/13	33/17	40/17	18	25/0	32/5.1
interval	17–26		25–36		24–31	25–40	36–45/15–19			
M3n	11/0.05	0	13/7.4	9.2	15/7.0	13/7.8	135/126	124	15/0	15/4.9
M1n	9.1	10	10	12	4.8	12	131	129	130	130
M1sn	3.5	3.5	8.2	9.0	4.7	8.9	131	129	77	111
M2n	19/0	0.20	15/5.9	9.2	13/0	16/9.1	133/128	126	325/0	325/4.0
M6	17/0.05	0	56/6.2	5.8	47/0	56/8.1	35/28	28	–	–

Over all seven characteristics; one number for rules that always decide. FA: safe distance on the false-accept side (failing alternatives). Interval: bootstrap 95 % interval of the decisive distance and, for a new family, of the safe distance (fits fixed). 126 cases per relation (interp.: 42 interpolated; extrap.: 84 extrapolated); a new family rests on two transfers and is scored in the floor of the produced family (Section 3.2). Pooled: the other geometry's alternatives added to the calibration set. Below the line, the further rules ~~–M6(M1sn: the force trend, the counterpart of M1s without the simulation) and, as a robustness variant, M6 (the M5 mean with a conformal margin on standardised leave-one-out residuals, appears only in the robustness analysis).~~

needs 4.5 and M5 9.7. Second, the extrapolation ~~eases are is~~ confounded with stroke speed: the ~~production-only rules lose almost only~~ nearest setting decides at 11 floors at 100 kN, where the press also ran slower ~~(the nearest setting decides at 11 floors~~, against 4.9 at 500 kN ~~), whereas the simulation-based interval rules are unsafe at both ends. Third, the production record of (the force trend at 8.9 at both). Taken together, the nearest setting decided at about 5 floors whenever the new design differed resolvably from the produced ones and force and speed did not change together, though each stratum has 34 to 58 cases and each distance rests on a handful of verdicts. Third, the other family's record does not help within a family with a produced sibling: pooling it makes most rules less decisive (M5 from 7.2 to 11 floors for a new variant, M2 from 23 to 53), M2n from 19 to 325), although it raises the coverage of though it raises M5's coverage from 86 to 90% and makes the centre of its interval the most accurate point prediction within the family (2.9 floors against 3.4 for the nearest setting).~~

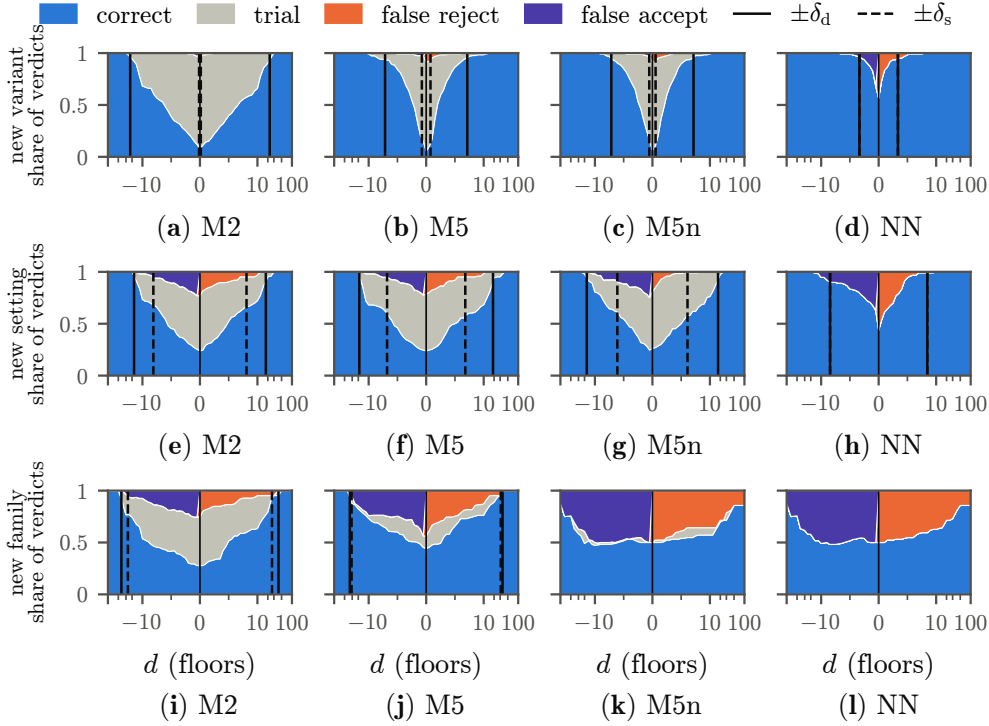


Fig. 2 Verdict shares against the requirement distance d (symmetric logarithmic axis, linear within ± 10 floors; 126 cases per panel; a new family in the produced family's floor), with the decisive and safe distances

The relation therefore does not govern decision fitness in general. Over the eleven calibrated rules and three relations, the relation explains 69.67 % of the variation of the log decisive distance and the rule 6 % , but this share comes from (69 and 6 % in the new family, where the 's own floor), but only because rules without the simulation fail for a new family by construction; over the two relations within a family, the rule explains 68 % and the relation 15 % . The (71 and 15 % with M1sn). Pooled over strata, the best attainable distance still grows with the novelty of the design, from grows with novelty: 3.4 floors with a sibling to 8.5 produced sibling, 8.2 for a new setting and 35.33 for a new family.

5.5 What the simulation and learning contribute

Pairing each calibrated rule with its counterpart without the simulation isolates what the simulation adds as it is used. Added as an offset, it adds nothing within a family with a produced sibling and degrades two rules: M5 decides where M5n does (difference 0.0 floors, -0.4 to $+1.1$), and while M2 and M1 decide 3.9 ($+0.5$ to $+7.4$) and 6.2 floors ($+4.6$ to $+9.7$) further out than M2n and M1n, and M1 13.5 further out for a new setting, because the simulated force effect is too large. Rescaled, it adds information within a family, where the scaled simulation decides 4.2 does no better: within a

geometry the nominal simulation is a function of the force, and the straight line in force (M1sn) decides 1.4 floors closer than the median of the produced alternatives (-5.4 to -0.5), though 1.5 floors less close than the nearest setting; scaled simulation with a sibling ($+0.65$ to $+3.0$) and 11 closer for a new setting the resealed trend is worse than none ($+5.3$ to $+13.8$), in every resample and refit configuration; the 4.2 floors that the scaled simulation gains over the median of the produced alternatives come from the force level. Only for a new family is the simulation decisive, by about 94–92 to 98 floors, and even there production knowledge carries over where a drawing nominal transfers: for the wall angle, the design angle plus the produced family’s deviation from its own design angle decides at 2.0 floors, against 7.3–7.5 for M5–A (for the arm angle and the dome, whose nominal is zero in both families, this baseline is M1n, at 44 and 27 floors). A Gaussian-process surrogate of the simulations reproduces them to within 1.8 over force, friction and thickness, with a leave-one-out root-mean-square error below 2 floors for 12 of the 14 characteristics and geometries, yet its decision fitness is that of the simulation it reproduces (decides as the simulation does (109 against 108 floors as M0, 15 against 15 as M1 with a sibling)).

Learning adds intervals rather than accuracy. Used as point rules, the centres of the M5, M5n and M3n intervals decide at 3.3, 3.1 and 3.7 floors within the family with a produced sibling, as close as the nearest setting, so the ranking of the learned rules rests on the width of their intervals. With the noise floor set by the variation between runs, (Table A3). With a sibling the Gaussian-process intervals cover 85–86 % of the true q_{95} within the family against a nominal 90 %, but their noise variance sits at its lower bound, the variance between the calibration series of one setting, in 97–99 % of these fits and in every other fit, so this coverage comes from the quasi-replicate series rather than from the learner; the largest-residual margin covers 89 % (it guarantees at least $(n-1)/(n+1) = 7/9$ under exchangeability, which these cases lack) and the jackknife+ interval of M3 intervals 96 % for (M3) and 89 % (M3n), above their guarantee of 7/9. For a new setting the calibrated intervals cover 43–55 %, and (M2n and M5n 93 % interpolating), for a new family the Gaussian processes at most 12 %. The inductive biases explain much of this: trees do not extrapolate, two force levels identify at most a linear trend, and in the median new-setting fit the length scale of the force sits at its lower bound, so the Gaussian process reverts to its mean.

5.6 Calibration budget

Figure 3 shows how the distances depend on the number of produced alternatives of the family and on their relation to the new design. A single produced sibling decides (Fig. 3): with one alternative at the new design’s force, the bias correction, the envelope rule and the nearest setting coincide by construction (the simulation terms cancel and a single residual gives no margin) and decide at 3.5 floors (2.3–4.6) without a wrong verdict at 10 floors. Further alternatives do not improve the nearest setting; they make the bias correction worse (12–18 floors), the envelope rule safer but less decisive, and the Gaussian processes M5, shown from three alternatives because a Gaussian process fits its hyperparameters to them, safer and more decisive (from 12–14 floors with two alternatives 13 to 7.2 with eight floors). Without a sibling, bracketing the new force buys safety but not decisiveness: from two alternatives on, the interval

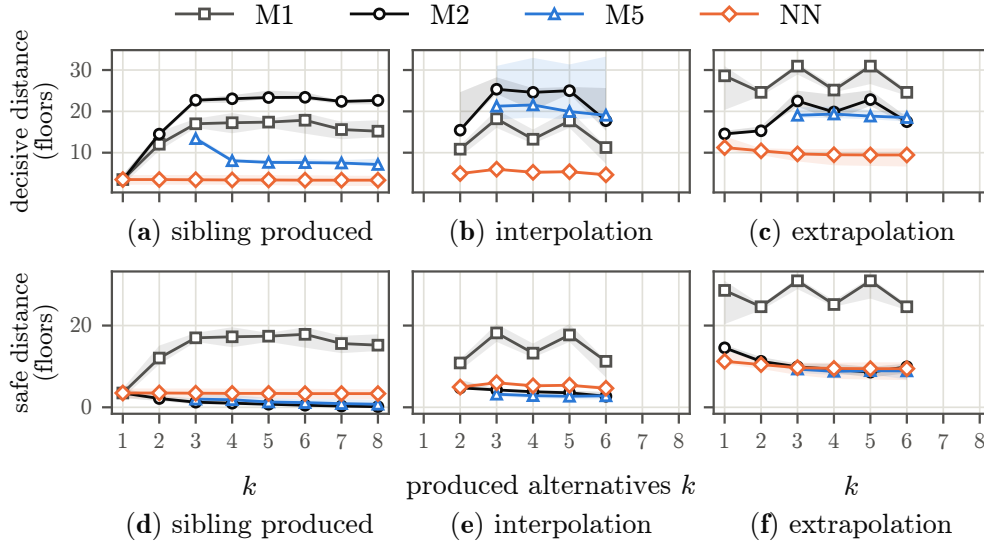


Fig. 3 Decisive and safe distances against the number of produced alternatives of the family, by the relation of the new design to them; bands: 95 % intervals over the held-out alternatives

rules the envelope rule and M5 are safe at 0–42.7–4.8 floors with no wrong verdict at 10 floors but decide only at 15–25 floors, against 4.7–6.0 for the nearest setting. Extrapolation, and extrapolation stays unsafe at every budget. Averaged over all subsets, the distances shrink with the budget mainly because the share of subsets containing a sibling grows, from 46 % with two alternatives to 96 % with six; the odd-even pattern of the bias correction comes from its median offset, which averages two forces only for an even number of alternatives. With calibration series of 50, 100 or 250 parts instead of 500, no distance within a family moves by more than 1.2–1.5 floors, and for a new family by at most 4 floors (Table A2): the qualification needs short runs of several variants rules can be calibrated on short runs, here scored against the truths and floors of the full series, and only the floor needs a run long enough to show the drift.

5.7 The procedure's decision rule

What a designer observes for a single verdict is the margin between the predicted interval and the requirement, and its value as evidence depends on the relation (Fig. 4): for a new variant with a produced sibling at least 95 % of the decided verdicts of the interval rules are correct at every margin; for a new setting the share grows with the margin, for the nearest setting from 85 % to 95 % beyond 5 floors; and for a new family M2 and M5 is right in 71–74 are right in at most 79 % of its decided verdicts whatever the margin their decided verdicts.

Step 5 turns this into guard bands: none for (Table A4). Under the reference prior the interval rules within the family and 0.25 floors for need none with a sibling and the nearest setting 0.25 floors; 2.0 floors for an interpolated setting, with which the

nearest setting needs 2.0 floors and then decides at 6.7 floors ~~is safe at 2.7 and sends~~
~~no design to trial~~ without trials at 10 floors; ~~7.0 floors~~ floors, whereas M2 and M5
 need 0.75 and 0.5 floors but send 31 and 40 % to trial; for an extrapolated ~~one, with~~
~~which it decides at 17~~ setting it needs 7.0 floors and sends a quarter ~~of the designs to~~
~~trial at 10 floors; and~~ to trial. M0 and M0w never qualify, nor do M1 and Mc with a
 sibling, M1s for an interpolated setting, M1, M1s, M2, M5 and Mc for an extrapolated
 one, or any rule for a new family ~~no guard band up to 20 floors reaches 95 %, so every~~.

These bands belong to the prior as much as to the rules. A prior concentrated
 near the truth, where release-level requirements lie (the grid within ± 5 floors), raises
 the band of the nearest setting to 2.0 floors with a sibling, 4.25 interpolating and
 14.25 extrapolating; a uniform prior within ± 20 floors lowers them to 0, 0.25 and 3.5
 floors and qualifies M2 and M5 for an extrapolated setting (6.0 and 5.0 floors), and
 one within ± 50 floors qualifies M2, M3 and M5 for a new family (6.25, 6.5 and 15
 floors). That every new-family design goes to trial ~~The guard~~ thus follows from the
 reference prior and the 20-floor cap, as the practice of trying out every new die would
 have it. The bands are chosen on the evaluation data, which makes the trial rates
 optimistic. ~~For a new setting the range guard refuses exactly the extrapolated forces,~~
~~after which M2, M5 and the nearest setting make no wrong verdict at 10 floors; the~~
~~novelty guard also refuses many cases that~~ and the guards refuse indiscriminately: 96 %
 of the range guard's refusals for a new setting would have been ~~decided correctly~~ right
 for the nearest setting and 72 % for M5 (novelty guard 96 and 73 %).

A worked example. A tryout engineer has to decide whether the concave cup at
 300 kN with medium lubrication, a new ~~process~~-setting, can be released without a trial
 run, with the six alternatives at 100 and 500 kN produced and a requirement, chosen
for illustration, that 95 % of the parts have a mid-side draw-in, by which tryout sets
the blank-holder force, of at most 15.30 mm. The floor is 0.050 mm, the relation is
 interpolation and no guard fires. ~~The Alone, the~~ nominal simulation predicts 15.66 mm
 and fails the design. The nearest produced setting, the mean of the six alternatives
at the neighbouring forces, predicts 14.99 mm, a margin of 6.2 floors, more than its
 guard band of 2.0 floors (5.0 under the narrowest prior), so step 6 returns meets;
~~beyond a margin of 6 floors every decided verdict of this rule for an interpolated~~
~~setting was correct.~~ Widened by their guard bands, M2, M5 and M5n return intervals
~~reaching 15.37, 15.58 and 15.33~~ 14.41–15.41, 14.23–15.61 and 14.66–15.33 mm and send
 the design to trial. The 500 parts had a 95th percentile of 14.88 mm, 8.4 floors below
 the limit: ~~the procedure was right, the simulation wrong and the interval rules cautious.~~
The case is typical, since at 8.5 floors above the truth step 6 with the nearest setting
is right in all 42 interpolated cases, but its guard band was estimated on data that
include it.

5.8 Requirements anchored in capability and in tolerances

Upper limits at which an alternative has a performance index of 1.0, 1.33, 1.67 or
 2.0 lie a median 1.0, 1.7, 2.5 and 3.1 floors above its 95th percentile, inside every
 decisive distance ~~except a near-replicate sibling's (0.85 floors).~~ The test is one-sided
by construction: every alternative meets ~~them~~ these limits, so a verdict ~~is~~ can only be
 right, a false reject or a trial, and a rule biased towards meets scores well. At an index

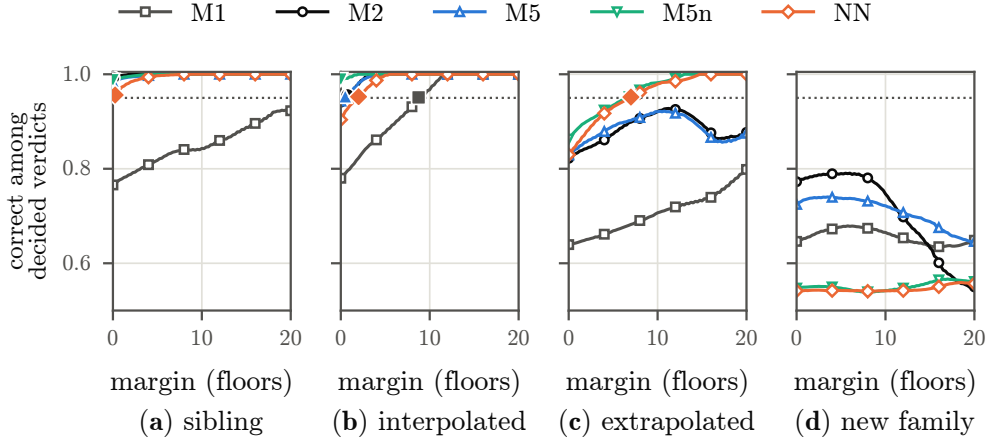


Fig. 4 Share of correct verdicts among the decided ones with at least the given margin between interval and requirement, under the reference prior (47 grid distances within ± 20 floors, equal weights); filled markers: guard bands

of 1.33 the nearest setting still says meets in 91% of the new-variant cases, because most of its errors are far smaller than its 95% distance, whereas M5 says so in 44% and sends the rest to trial; shares of right verdicts, false rejects and trials are, with a produced sibling and for a new setting the two are right in, 91/9/0 and 65/35/0% for the nearest setting, 44/2/53 and 32/16/52%, for M5, and 81/19/0 and 79/21/0% for the scaled simulation, which falsely accepts failing alternatives elsewhere; for a new family no rule exceeds is right in more than 54%. Release-level requirements are therefore decided thus decided reliably at the design stage only within a family with a produced sibling, and then by the production record. General angular tolerances of ISO 2768-1 (ISO, 1989) give a coarser scenario that mostly tests whether a rule recognises gross failures (only 32 of its 108 decisions lie within 5 floors of the limit), 6 of them on failing alternatives, and a cost comparison inherits this coarseness. With a false reject at half the cost of a false accept, its the cheapest rule is (i) with a produced sibling, the nearest setting at any cost for a new variant; the scaled simulation and the force trend, tied at zero cost; (ii) for a new setting, the production-only interval rules while a trial costs at most a tenth of a false accept, and the nearest setting beyond; force trend at every trial cost; and (iii) for a new family, a trial of every design while a trial costs at most a tenth of a false accept, the envelope rule at a fifth and M5 beyond.

5.9 Robustness

The analysis was repeated under variants of the data, the constants and the model specifications (Table A2). Adjusting for punch temperature changes no calibrated distance by more than 2.6 floors. Dropping the warm-up of the first 150 parts shrinks the floors and so lengthens the distances (the nearest setting 4.1 floors with a sibling, 12 for a new setting), and floors taken between series, 1.5 to 9.1 times the reference, shorten them (0.75 and 2.7 floors) shorten them. Other floor protocols rescale the decisive distances by 0.60 to 1.42, and the Gaussian-process specifications move them by at most

2.1 floors, ~~and in the unit of the reproducibility of q_{95} the nearest setting decides at 2.8 and 6.9 floors.~~ In every variant that includes it, ~~the nearest setting remains the them, the~~ most decisive rule within a family ~~and the envelope rule the safest for a new family~~ is the nearest setting or the force trend, and with a produced sibling the nearest setting (tied once). The refitting bootstrap widens the intervals of the learned rules most (M5 4.0–11 floors ~~for a new variant~~ with a sibling, the nearest setting 2.3–4.6), yet in all 200 resamples (48 configurations) the nearest setting is ~~the more decisive in all 200 resamples, against more~~ decisive than M5 and M5n, with a sibling and for a new setting alike, and the force trend than the scaled simulation (by 1.35–3.6 and 8.8–12 floors); point estimates at the edge of their refit intervals (M1s with a sibling at 4.9 floors, 4.9–7.3) reflect what losing a pattern does.

6 Discussion

6.1 What the evaluation establishes

Five findings answer the research questions. Two follow from the definitions and hold wherever the framework is applied: the floor puts every rule and characteristic on a common production-referenced scale, though not on a common scale of nonconforming fractions, and as a within-run quantity it ~~is a lower bound of bounds~~ the variation of series production from below (RQ1); and an interval rule trades decisiveness for safety, which the two distances make explicit, ~~and reaches its nominal while its distribution-free intervals are guaranteed their~~ level only where the calibration alternatives are exchangeable with the new design. Three are observations on this dataset and hypotheses for confirmation: the best attainable distance grows with the novelty of the design, from 3.4 floors with a produced sibling (0.85 for near-replicates, 4.9 when both siblings differed resolvably) to 8.2 for a new setting (4.7 interpolated; 4.9 and 8.9 extrapolated to 500 and 100 kN) and 33 for a new family, while within a family the choice of rule matters more than the relation (RQ2); production evidence buys safety before decisiveness, ~~since interval rules become safe with a few produced alternatives that bracket the new design and decisive only with near-replicates~~ and interval rules held their level with a produced sibling but not when extrapolating or for a new family; and with this simulator and six to nine produced ~~settings per fit~~ alternatives at two or three force levels, the simulation ~~contributed~~, as an offset or rescaled, contributed only across families, ~~a fitted scale made it useful within a family but harmful for a new setting,~~ and learned models predicted as well as the nearest produced setting with a produced sibling while adding abstention (RQ3).

6.2 Using the framework in design

As a planning heuristic (Table 5 ~~separates guidance that follows from the framework from observations on this dataset. Before a requirement is fixed,~~ the expected margin between ~~requirement and capability~~ a requirement and the predicted 95th percentile can be compared with the safe distance of the best rule qualified for the relation ~~of the design; if the margin;~~ if it is smaller, a trial or ~~further more~~ production evidence has to be planned, and the decision itself needs the guard band, because the safe distance

Table 5 Guidance for design-stage manufacturability decisions, with its basis

Situation	Guidance	Basis
<i>From the framework (prescriptive; usefulness not evaluated)</i>		
Before relying on a predictor	Measure the floor on consecutive production of the family with a stated protocol and report σ_b , σ_w and its lag dependence; treat it as a lower bound of series variation.	framework
Qualifying a rule	Evaluate it for the evidence relation of the coming designs by grouped cross-validation, and split near-replicate siblings from resolvable ones.	framework
Choosing a rule	Compare it with the nearest produced setting and with its own version without the simulation, on decisive and safe distance jointly or by expected cost.	framework
Deciding a single design	Derive the rule's guard band for the distribution of requirement distances the team expects and apply it to the observable margin; a margin inside it, or a fired guard, means a trial.	framework
Planning a requirement	As a heuristic, plan a trial if the expected margin between requirement and predicted 95th percentile is below the safe distance of the best qualified rule; no decision is reliable closer than a trial reproduces the quantile.	framework
<i>Observed on this dataset (18 alternatives, 2 geometries, 7 characteristics)</i>		
New variant with a produced sibling	nearest produced setting right beyond 3.4 floors (0.85 for near-replicates, 4.9 for resolvable siblings); interval rules safe below one floor, coverage 85–96 %	126 cases
New process setting	nearest setting or force trend right beyond 4.7 floors interpolated, 4.9 extrapolated to 500 kN and 8.9 to 100 kN (with a speed change); interval rules safe at 5.9–16 floors, coverage 43–55 %	42 + 84 cases
New family	best rule safe at 16–17 floors in either family's floor, which rule not resolved; rules without the simulation fail; a transferable drawing nominal helps	two transfers
Release-level requirements	lie 1–3 floors above the 95th percentile, inside every decisive distance except a near-replicate's; decided reliably only with a produced sibling	126 cases

is defined on the true margin. Release-level requirements make this the usual case: they lie within a few floors of the capability, where only a near-replicate production record decides reliably, consistent with, as the practice of requiring a production trial run for a process change. For a single design, step 6 gives a verdict whose guard band a designer can see, and the worked example shows that the most decisive rule is also the most transparent: the nearest produced setting points to the produced alternative it copies, whereas a Gaussian-process interval does not. Which rule to prefer depends on the costs, which differ by relation: a trial of expects: the rules decide reliably where a trial is cheapest, a new force on an existing tool is being a short press run, a trial of and none does near release-level limits where a trial is dearest, for a new family that may need a new tool. In a company, The most decisive rule is arguably also the most transparent: the nearest produced setting points to the produced alternatives whose record it averages. The variants that steps 1 to 5 fit the start of production or the need come from process development and tryout, whose short series of several settings support the sibling and new-setting relations, and from a family's production history across part numbers, which supports the new-family relation; production part

1059 approval of a family, when its series are measured anyway, and step 6 the later decisions
1060 on variants and process changes, whose initial process study covers the released setting,
1061 supplies the floor but not the variants.

1062

1063 6.3 Simulation and machine learning in this regime

1064 Within a design family the production record carried the decision, even against a
1065 rescaled simulation. The simulated force trend of the two draw-ins was 2.7 to 5.1 times
1066 too large, and calibrating the friction coefficient did not repair it, which points to the
1067 friction law and the blank-holder model rather than to a parameter. ~~Learned models~~
1068 ~~added an interval that is safe where the calibration alternatives are exchangeable with~~
1069 ~~the new design and over-confident where they are not.~~ A multi-task model cannot
1070 help a family with no produced alternative, because the correlation between families
1071 cannot be estimated without its data; ~~the pooled Gaussian process, which carries~~
1072 ~~the geometry as an input, is the borrowing of strength the data allow, and it in~~
1073 ~~the pooled scopes the Gaussian process with a 0/1 geometry input is a restricted~~
1074 ~~intrinsic coregionalisation model (equal variances, one positive correlation between~~
1075 ~~the families, shared length scales), which improved coverage but not decisiveness.~~
1076 These findings concern one simulator of low fidelity ~~for this production~~, used through
1077 offsets and one scale, and learners with six to nine produced ~~settings each; a surrogate~~
1078 ~~of these simulations inherits the simulator's distance, whatever its accuracy against~~
1079 ~~the simulation. For the evaluation of alternatives at two or three force levels. For~~
1080 ~~evaluating~~ AI-based DFM support the lesson is methodological: ~~held-out accuracy is~~
1081 ~~not enough;~~ a learned model should be compared with a production-record baseline,
1082 scored by evidence relation in a unit that production defines, and shown to its user
1083 with a guard band. ~~Whether that makes designers' reliance match a rule's reliability~~
1084 ~~(Lee and See, 2004), and when an abstention should go to an expert rather than to~~
1085 ~~a trial (Madras et al, 2018), are hypotheses for a study with design~~ A study with
1086 ~~design teams could test two hypotheses on reliance (Lee and See, 2004), with cases of~~
1087 ~~known production outcome: designers shown the guard-banded verdict accept fewer~~
1088 ~~wrong verdicts within the safe distance than designers shown the verdict alone, at an~~
1089 ~~equal rate of right acceptances; and, as the same AI assistance helped some teams and~~
1090 ~~hindered others (Zhang et al, 2021), the guard band narrows the spread of decision~~
1091 ~~quality between~~ teams.

1092

1093 6.4 Implications for design research

1094 The ~~framework gives several results give~~ constructs of design research a
1095 production-referenced measure. For variant ~~, adaptive~~ and platform design (Pahl
1096 et al, 2007; Simpson et al, 2006) ~~, and the capability databases that serve~~
1097 ~~them (Tata and Thornton, 1999; Thornton, 2004), the results argue for indexing~~
1098 ~~production records by their relation to the distances by evidence relation~~
1099 ~~quantify how far production knowledge carries from produced variants to a~~
1100 ~~new one, with the nearest produced setting as the baseline that any capability~~
1101 ~~database (Thornton, 2004) or learned model has to beat~~ coming design rather
1102 ~~than for a more sophisticated model on top.~~ The safe distance is the part
1103
1104

Table 6 What the framework requires, mapped onto other processes (an untested proposal)

Process	Batch and run	Main variation	Evidence relations	Floor
Sheet forming (this paper)	consecutive parts of one series	drift in a run; coil, die state	variant, setting, family	batch centres in a run
Injection moulding	shots of one cavity and set-up	cavity, shot, set-up, material lot	cavity, process window, new mould	per cavity; cavity offsets as effects
Machining	consecutive parts between offsets	tool wear, offset-compensated	tool path, cutting data, new part	batch centres within an offset interval
Casting	castings of one pattern and melt	melt, pattern position	alloy, gating, new pattern	within a melt; melts as runs
Additive manufacturing	replicates in one build	build, position, orientation	orientation, settings, new part	within a build, via replicated anchors

of a margin that covers predictive uncertainty, as distinct from the part that absorbs change (Eckert and Isaksson, 2017; Eckert et al, 2019). ~~For testing strategies (Thomke and Bell, 2001; Salado and Kannan, 2018; Camburn et al, 2017), overlapped development (Krishnan et al, 1997; Terwiesch et al, 2002) and The growth of the decisive distance from about 3 to 8 and over 30 floors says for which overlaps of development (Krishnan et al, 1997; Terwiesch et al, 2002) preliminary manufacturability information can be released, the new-family safe distance, some 0.8 mm of draw-in, how far a set-based design (Sobek et al, 1999), the distances say can narrow a feasible set before production evidence exists, and the distances and trial rates when a test is cheaper than a decision, when design-stage information is fit to act on and how wide a feasibility limit has to be (Thomke and Bell, 2001; Salado and Kannan, 2018).~~ For validation, the evidence relation is the question of the application domain (Oberkampf and Roy, 2010); ~~and the distances could serve as evidence of a risk-informed credibility assessment (ASME, 2018).~~ In the terms of the design research methodology (Blessing and Chakrabarti, 2009), the evaluation ~~is a support evaluation: it~~ shows structural validity ~~in the sense of the validation square,~~ that the metrics discriminate between rules and relations on real data (Seepersad et al, 2006); ~~and is a first step towards performance validity, which would require showing that decisions taken; performance validity would require better decisions~~ with the framework ~~are better than decisions taken without it.~~

6.5 Reuse for other processes and AI-based DFM tools

The framework needs several produced variants of a design family whose relations ~~to each other~~ span those of the coming designs, a series of consecutive parts or replicated batches of one design per variant, a design-stage predictor for every variant, and characteristics measured identically on parts and predictions. ~~Here the long series served the floor, whereas calibration runs of; here rules calibrated on~~ about 50 parts per variant ~~qualified the rules scored~~ as well as ~~full series; how many long runs a floor needs was not tested. Table 6 maps these requirements onto the processes named most often in DFM; attribute with full series, and only the floor needed long runs.~~ Attribute outcomes such as splits, ~~porosity or short shots~~ need a requirement on a rate ~~rather than a quantile,~~ which the framework does not yet cover. ~~A predictor~~

1151 ~~need not return an interval: any tool that returns meets, fails or trial for a stated~~
1152 ~~requirement, such as a classifier at its operating threshold or a~~ Other predictors need
1153 three adaptations. A classifier with a fixed threshold does not take the requirement
1154 as input, so its operating characteristic is estimated by binning test cases by their
1155 true margin, not by the exact estimator of Section 3.4, unless the requirement is made
1156 an input; a probabilistic classifier needs two thresholds to return a trial. A language-
1157 ~~model check given the requirement value, can be scored by sweeping the requirement,~~
1158 may return verdicts that are not monotone in the requirement and that vary between
1159 queries, so κ and ω are computed on a requirement grid with repeated queries,
1160 and the share of non-monotone verdict sequences is reported. For continuous design
1161 spaces the relation becomes a distance from the produced designs in descriptor space
1162 or the position relative to their hull, and each stratum needs enough cases: with about
1163 240 verdicts per relation, the 95 % threshold already rests on about a dozen. For a
1164 generative tool, its manufacturability check is scored. A minimal report comprises

- 1165 1. ~~the floor with its protocol and variance components, the,~~ variance components and
1166 lag dependence;
- 1167 2. decisive and safe distances by relation~~with refitting,~~ with refit intervals and one-
1168 sided values,~~the coverage;~~
- 1169 3. coverage and half-width of interval rules, and the source of their noise variance;
- 1170 4. the nearest-produced-setting and no-simulation baselines, and;
- 1171 5. the guard bands ~~and guard performance. Distances per characteristic rest on as~~
1172 ~~many cases as there are produced alternatives, so a team would pool characteristics~~
1173 ~~with similar ratios of floor to part scatter, or accumulate evaluations over~~
1174 ~~families with their prior, the trial rates and the performance of the guards;~~
- 1175 6. for language-model tools, the prompt and the query protocol.

1177 Table 6, an untested proposal, maps these requirements onto other processes; distances
1178 in floors are comparable within a protocol, not across processes, since a floor between
1179 builds, cavities or offset intervals is a between-unit quantity (here the floor between
1180 series is 1.5 to 9.1 times that within a run). An attempt ~~to apply the framework to an~~
1181 ~~open dataset of on open~~ powder bed fusion data of PA12 (Leirimo, 2021) failed on the
1182 second requirement: ~~grouping specimens by orientation class orientation classes~~ mixed
1183 orientations in a batch, ~~and eight identical anchor specimens eight identical anchors~~
1184 per build were too few for a floor ~~; a between-build floor from replicated anchors would~~
1185 ~~be the remedy within a build, and three builds give three differences between builds.~~

1187 6.6 Threats to validity

1188 The evaluation covers one material, one press and two geometries, ~~and the with~~ rela-
1189 tions within a family ~~come~~ from a 3×3 factorial in which lubrication is mostly not
1190 resolved; the procedure is general, the numbers are not. The new family rests on two
1191 transfers between geometries that differ in three features at once, and the intervals of
1192 its distances hold the two fits fixed. Each alternative was produced once, so the floor
1193 covers drift within a run but not between days, coils or tool states, the reference q_{95}
1194 is that of one run, and the order and dates of the series are unknown; between-series
1195 variation bounds the missing component at 1.2–9.8 floors ~~(Section 5.1). Scanner, and~~

scanner drift cannot be separated from process drift. The extrapolation ~~in force to~~ 100 kN is confounded with stroke speed, which the simulation ~~does not represent, so~~ the extrapolation results describe a force and speed change together ~~omits~~. The requirements are one-sided limits on the 95th percentile of a characteristic averaged over sides, below the conformance of series release. Several rules, the grouped schemes ~~and~~ the guard bands ~~and their prior~~ were added or chosen after earlier results had been seen and are evaluated on the same data, so the comparison between rules carries selection optimism (Cawley and Talbot, 2010); ~~the analysis was carried out by~~ a single analyst carried out the analysis, with code and data open for replication.

7 Conclusions

Design-stage manufacturability decisions can be evaluated against production on a scale ~~that~~ production defines: the decisive and safe distances in floors tell how close to a requirement a rule decides and how close it can be relied on not to mislead, and a guard band derived for the requirements a team expects turns them into a rule for ~~a single decision.~~

single decisions. On 18 deep-drawn ~~design process~~ alternatives with matched simulations, the best attainable distance grew with ~~the novelty of the design novelty~~: 3.4 floors with a produced sibling ~~7.5 (0.85 for near-replicates, 4.9 for resolvable siblings)~~, 8.2 for a new process setting (4.7 interpolated; 4.9 extrapolated to 500 kN and 8.9 to 100 kN, with a stroke-speed change), and no rule safe closer than ~~16–16–17~~ floors for a new family, in either family's floor. Within a family the choice of rule mattered more than the relation; release-level requirements lay inside every decisive distance ~~except~~ a near- replicate's and need a trial unless a sibling has been produced; and in this small-data regime the simulation ~~as used, a model with a large and partly sign-inconsistent bias, offset or rescaled~~, contributed only across families, while learned ~~models added abstention rather than accuracy~~ 90 % intervals covered 85–96 % of the truths with a produced sibling but under 50 % extrapolating or for a new family.

Four extensions follow: series ~~produced on from~~ several days, coils and tool states; ~~so that the floor covers variation between runs; design~~; families with continuous geometric descriptors; a confirmatory evaluation test of the frozen procedure on ~~another dataset or process~~ other data; and a study ~~with design teams of whether decisions taken with the framework are better~~. The released code, which reproduces every result from the public data, ~~is the starting point of whether design teams decide better with it~~.

Appendix A Notation, further results and computation details

Table A1 summarises the notation of Section 3, ~~and~~ Table A2 gives the distances under the robustness variants; ~~the other results quoted in the text, such as guard bands, coverage, paired differences and distances per characteristic, are written by the released code and archived with it~~ Table A3 the coverage, width and centre of the interval rules, and Table A4 the procedure's decision rule by relation. Table A5 traces

every other number quoted in the text to the script that computes it and the result file that holds it in the archived release (Canseven, 2026).

Computation details. Batches need at least 80 % valid values. M5 and M5n are Gaussian-process regressions (constant times anisotropic squared-exponential kernel plus white noise, normalised targets, [hyperparameters by maximum marginal likelihood without hyperpriors](#), three restarts, length scales in $[0.1, 100]$) on a geometry indicator, the force in 100 kN and the lubrication rank; [the lower bound of the noise variance, the larger of the mean sampling variance of the calibration alternatives' \$q_{95}\$ and the pooled variance between their series of one geometry and force, is expressed on the normalised scale](#). NN takes the mean q_{95} at the nearest force of the family, both neighbours when equally near; [M1sn fits \$q_{95} = a + b\$ force by least squares](#). Mc calibrates, per lubrication pattern, the friction coefficient whose simulated draw-ins are closest to the calibration centres. M3 and M3n use histogram gradient boosting (depth 3, 200 iterations, rate 0.08) on geometry, force, lubrication, sheet thickness and oil film, with 500 incoming conditions drawn from calibration parts of the same pattern; their jackknife+ interval ~~combines~~ [takes](#) the leave-one-out predictions for the new alternative ~~with the absolute~~, [shifted by the median](#) leave-one-out ~~residuals about their median residual, minus and plus the absolute residuals about that median, with its bounds at the extreme order statistics when n is too small for the 0.9 quantiles. Guard bands lie on a grid of 0.25 floors. The refitting bootstrap draws the three lubrication patterns of each geometry with replacement 200 times, at least two of them distinct, [and does not restrict the noise floor to distinct series](#). Software: Python 3.11 with NumPy, pandas, SciPy, scikit-learn (Pedregosa et al, 2011) and Matplotlib.~~

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Declarations

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Data availability. All data are public. RDDAC: doi:10.18419/DARUS-5589 (version 2.0); DDACS: doi:10.18419/DARUS-4801 (version 3.0); both licensed CC BY 4.0. The extracted per-part and per-simulation feature tables ~~are provided and all result files are archived~~ with the code [\(Canseven, 2026\)](#).

Code availability. The complete analysis pipeline ~~is released under an open licence at (directory scripts/) and~~, [which](#) reproduces every number in this paper from the public data; ~~a versioned archive of the code and~~, [is archived with](#) the feature

⁰Rules refitted for every variant; empty cells: the variant does not affect the rule. Floor variants and the unit of q_{95} reproducibility (95 % quantile of the difference between the q_{95} of two batches of 100 parts) rescale the distances. Calibration series of n : the calibration alternatives' q_{95} from their first n parts. Conformance p : the decided statistic is the p -quantile. Response surface: nominal simulation and envelope from a quadratic fit over thickness and friction. Other kernels and descriptors of the GPs move M5 by at most 0.4 floors. Refit bootstrap: 95 % interval when the lubrication patterns of each geometry are resampled and every rule is refitted (200 resamples).

Table A1 Notation and terms

Symbol, term	Meaning
$a, c, q_p(a, c)$	alternative (design–process combination), characteristic, p -quantile of the parts of a ($p = 0.95$)
R, d	requirement (upper limit on c or $ c $); its distance from the truth in floors, $R = q_p + dF_c$
$[L_r, U_r], V_r$	interval of rule r for q_p ; verdict meets, fails or trial (Eq. 1)
B, m, F_c	batch size (50); batch centre (10 % trimmed mean); production floor (Eq. 2)
σ_b, σ_w	standard deviation between batch means and between parts within a batch
ESR, MRC	effect-to-scatter ratio (Eq. 3); minimum resolvable change, $F_c/ s_c $
u, v	distances in floors by which an interval excludes the truth from above and below
$\kappa(\delta), \omega(\delta)$	shares of correct and wrong verdicts among the feasible verdicts at $ d = \delta$
δ_d, δ_s	decisive distance ($\kappa \geq 0.95$ beyond it) and safe distance ($\omega \leq 0.05$ beyond it)
FA, FR side	one-sided distances for failing alternatives (false accepts) and adequate ones (false rejects)
g	guard band: observable margin beyond which at least 95 % of the decided verdicts were correct, for a stated prior of requirement distances, here the 47 grid distances within ± 20 floors with equal weights (Section 3.6); not a property of rule and relation alone
Calibration alternatives	produced alternatives whose record calibrates a rule
Evidence relation	sibling (same setting produced), new process setting (interpolated, extrapolated), new family
Calibration budget	distances against the number k of produced alternatives, by relation
Guard	family, range and novelty checks that send a design outside the calibration to trial
Measurement resolution	step between adjacent values that the measuring system reports (gauge sense)

tables ~~is deposited on Zenodo~~ [\[DOI pending\]](https://doi.org/10.5281/zenodo.1234567) and result files of this version on Zenodo (Canseven, 2026); Table A5 maps the numbers to its scripts and files. A working copy is at <https://github.com/htcanseven/Claude> (directory [scripts/](#)).

Author contributions. H.T.C. conceived the method, designed and carried out the analysis, and wrote the paper.

Use of generative AI. A large language model (Claude, Anthropic) was used under the author’s direction for code development, literature retrieval, data analysis, drafting and editing, and simulated peer review of earlier versions. The author designed the study, verified all analyses, checked every reference against its published record, and takes full responsibility for the content. All figures are produced by the released code from the data; no AI-generated images are used.

Table A2 Decisive/safe distances (floors) under the robustness variants

Variant	New variant				New setting			New family		
	M1s	M2	M5	NN	M2	M5	NN	M1sn	M2	M5
Reference	4.9	23/0.15	7.2/0.75	3.4	18/8.1	19/6.8	8.5	8.2	51/16	38/28
Temperature-adjusted	5.0	23/0.10	7.2/0.65	3.4	17/7.6	19/6.8	9.4	8.9	48/15	37/27
Warm-up dropped	6.5	27/0.20	9.0/0.45	4.1	21/8.7	24/8.7	12	8.8	63/18	39/29
Floor between series	2.9	11/0	1.3/0.20	0.75	9.9/2.6	11/2.3	2.7	4.5	16/3.8	12/9.3
Floor: batches of 25	4.6	21/0.15	6.4/0.70	2.9	16/6.9	17/6.3	7.7	7.3	43/14	31/23
Floor: batches of 100	5.3	24/0.15	8.2/0.80	3.5	19/8.5	21/7.9	8.9	8.9	57/17	46/29
Floor: 90 % quantile	6.2	31/0.20	8.8/0.90	4.0	21/10	23/8.6	10	10	61/20	45/34
Floor: 99 % quantile	3.4	14/0.10	5.2/0.50	2.4	11/4.4	13/4.4	5.7	6.1	37/11	29/15
Calibration: 50 parts	5.0	22/0.20	6.8/0.70	3.4	18/7.5	19/6.2	8.4	7.7	50/16	37/28
Calibration: 250 parts	4.9	22/0.45	7.0/0.45	3.3	18/7.5	19/6.2	8.7	8.3	50/16	38/28
Conformance $p = 0.90$	4.9	23/0.20	7.2/0.70	3.3	18/7.5	18/6.4	8.4	8.1	51/15	38/28
Conformance $p = 0.99$	5.1	23/0	7.5/0.70	3.3	18/8.3	20/6.9	8.7	8.4	51/16	38/28
Interval level 0.80		23/0.15	6.4/1.0		18/8.1	17/8.1			50/17	38/28
Interval level 0.95		23/0.15	8.0/0.25		18/8.1	20/6.2			51/16	39/27
Response surface	4.8	25/0.30	7.2/0.70	3.4	18/7.6	20/6.9	8.5	8.2	49/25	37/27
GP: sampling bound			6.7/1.4			18/7.4				38/30
Unit: q_{95} floor	5.3	17/0.15	6.9/0.40	2.8	15/6.0	18/6.2	6.9	7.1	41/13	34/30
Refit, δ_d	4.9–7.3	22–23	4.0–11	2.3–4.6	16–18	18–20	6.9–9.2	7.3–10	49–53	34–39

Rules refitted for every variant; empty cells: the variant does not affect the rule; a new family in its own floor. The floor between series contains the lubrication effect and, for a new variant, the series that NN averages, which makes it close to circular there. Warm-up: first 150 parts. Calibration: calibration q_{95} from the first n parts, truths and floors from the full series. Response surface: simulations from a quadratic fit over thickness and friction; other GP kernels and descriptors move M5 by at most 0.4 floors. Unit: the q_{95} floor, 95 % quantile of the difference between the q_{95} of two batches of 100 parts. Refit: 95 % interval of the decisive distance over 200 draws of each geometry's lubrication patterns with replacement, every rule refitted (Section 4.4).

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Table A3 Interval rules by relation: coverage, half-width and the interval centre as a point rule

Rule	Sibling produced			Interpolated setting			Extrapolated setting			New family		
	Cov	HW	Ctr	Cov	HW	Ctr	Cov	HW	Ctr	Cov	HW	Ctr
M2	89	7.6	8.6	71	5.7	10	44	4.8	13	47	7.6	34
M2n	90	5.9	9.1	93	5.0	4.8	33	3.1	12	12	5.7	131
M3	96	9.3	9.9	43	7.0	19	43	8.9	21	40	9.7	29
M3n	89	3.0	3.7	36	3.4	11	48	2.3	9.9	7	3.3	130
M5	86	2.2	3.3	71	6.4	9.8	46	4.6	13	12	2.3	34
M5n	85	2.3	3.1	93	5.8	4.7	36	3.7	12	0	2.2	130

Cov: share of cases whose true q_{95} lies in the interval (%; nominal 90). HW: median half-width (floors). Ctr: decisive distance of the interval centre used as a point rule (floors). A new family in the floor of the produced family. The GP noise variance sits at its lower bound in 97 and 99 % of the M5 and M5n fits with a sibling and in every other fit.

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Table A4 The procedure's decision rule (step 6) by relation: guard band, distances and trial shares

Rule	Sibling produced				Interpolated setting				Extrapolated setting			
	g	δ_d/δ_s	T_{10}	T_{20}	g	δ_d/δ_s	T_{10}	T_{20}	g	δ_d/δ_s	T_{10}	T_{20}
M1	none	–	100	100	8.75	20/2.1	44	4	none	–	100	100
M1s	2.0	6.9/2.9	1	0	none	–	100	100	none	–	100	100
M2	0	23/0.15	33	8	0.75	19/2.0	31	4	none	–	100	100
M5	0	7.2/0.75	1	0	0.5	20/2.3	40	4	none	–	100	100
M5n	0	7.2/0.55	0	0	0	16/0	25	0	6.75	25/1.9	48	15
NN	0.25	3.6/3.1	0	0	2.0	6.7/2.7	0	0	7.0	17/2.5	26	1
M3	0	23/0	45	6	17.75	47/0	94	69	16.75	51/0.05	85	69
M3n	0	11/0.05	9	0	6.0	21/1.0	55	6	6.5	19/1.3	47	3
M1sn	0.75	4.3/2.8	0	0	2.0	6.7/2.7	0	0	6.0	15/2.9	21	0

g : guard band (floors) under the reference prior (Section 3.6); δ_d/δ_s : distances of the rule with its interval widened by g ; T_{10} , T_{20} : share of verdicts sent to trial (%) at requirements 10 and 20 floors from the truth. None: no band up to 20 floors reaches 95 %, so the procedure sends every design to trial; so also M0 and M0w in every relation, Mc with a sibling and for an extrapolated setting (for an interpolated one $g = 18$), and every rule for a new family, scored in the produced family's floor.

Table A5 Where the numbers quoted in the text come from: scripts and result files (results/*.csv)

Section	Results	Script: result files
3.4, 4.1, 4.3	failing-side weight, run metadata, scan-line scatter, measurement resolution	panel.py: panel.failing_weight, panel_runs, panel_resolution_steps; measurement.py: meas_scan
5.1	time order, lag profile, normal model, subgroups, part-to-part share, between series, q_{95} reproducibility, in-line drift	alternatives.py: alt_floor; sensitivity.py: sens_drift; panel.py: panel_variogram, panel_floor_protocol; measurement.py: meas_floor; robustness.py: rob_floor_between; inline.py: inline_drift
5.2	resolvable contrasts and changes, simulation offsets, force effect, arm sign, cup depth	alternatives.py: alt_effects, alt_mrc; robustness.py: rob_bhf_pairs; decisions.py: dec_alternatives; panel.py: panel_force_effect, panel_arm_sign, panel_m0_split
5.3–5.4	distances in both floors, paired differences, strata, force levels, decomposition	decisions.py: dec_resolution, dec_paired; panel.py: panel_source_floor(_paired), panel_sibling_strata, panel_force_levels, panel_decomposition
5.5	simulation pairs, drawing nominal, surrogate, coverage, GP fits	decisions.py: dec_paired, dec_coverage, dec_gp; panel.py: panel_nominal_offset; design_space.py: ds_surrogate(_rule)
5.6, 5.9	budget, short series, robustness variants, refit, q_{95} unit	budget.py: budget_design, budget_resolution; robustness.py: rob_decisions, rob_refit(_paired); panel.py: panel_q95_unit
5.7	margins, guard bands and priors, guards, worked example	panel.py: panel_step6(_curves), panel_guard_priors; guard.py: dec_guard; decisions.py: dec_intervals
5.8	capability-anchored limits, ISO 2768-1 scenario, costs	panel.py: panel_capability; scenarios.py: scen_strata, scen_scores, scen_cost_map
6.5	PA12 attempt	case2_pa12.py: case2_floor, case2_effects

scripts/run_all.sh runs the scripts in order; make_tables.py writes every table and figs.py every figure from the result files.

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